1.

```
[alvare12@elvis txtFiles]$ grep i paragraphs.txt
This file is a bunch of randomly generated paragraphs https://randomword.com/paragraph
He was an expert but not in a discipline that anyone could fully appreciate. He knew how
 to hold the cone just right so that the soft server ice-cream
fell into it at the precise angle to form a perfect cone each and every time. It had tak
en years to perfect and he could now do it without even putting
any thought behind it. Nobody seemed to fully understand the beauty of this accomplishme
nt except for the new worker who watched in amazement.
There was something special about this little creature. Donna couldn't quite pinpoint wh
at it was, but she knew with all her heart that it was true. It
wasn't a matter of if she was going to try and save it, but a matter of how she was goin
g to save it. She went back to the car to get a blanket and
If you can imagine a furry humanoid seven feet tall, with the face of an intelligent go
illa and the braincase of a man, you'll have a rough idea of
what they looked like -- except for their teeth. The canines would have fitted better in
 the face of a tiger, and showed at the corners of their wide,
thin-lipped mouths, giving them an expression of ferocity.
Where do they get a random paragraph?" he wondered as he clicked the generate button. Do
 they just write a random paragraph or do they get it
somewhere? At that moment he read the random paragraph and realized it was about random
paragraphs and his world would never be the same...
```

Its looking for any “i” not capital and highlighting it

2.

```
[alvare12@elvis txtFiles]$ grep -w the paragraphs.txt
He was an expert but not in a discipline that anyone could fully appreciate. He knew how
 to hold the cone just right so that the soft server ice-cream
fell into it at the precise angle to form a perfect cone each and every time. It had tak
en years to perfect and he could now do it without even putting
any thought behind it. Nobody seemed to fully understand the beauty of this accomplishme
nt except for the new worker who watched in amazement.
wasn't a matter of if she was going to try and save it, but a matter of how she was goin
g to save it. She went back to the car to get a blanket and
when she returned the creature was gone.
If you can imagine a furry humanoid seven feet tall, with the face of an intelligent gor
illa and the braincase of a man, you'll have a rough idea of
what they looked like -- except for their teeth. The canines would have fitted better in
 the face of a tiger, and showed at the corners of their wide,
Where do they get a random paragraph?" he wondered as he clicked the generate button. Do
 they just write a random paragraph or do they get it
somewhere? At that moment he read the random paragraph and realized it was about random
paragraphs and his world would never be the same...
```

3.

```
[alvare12@elvis txtFiles]$ sed 's/Dracula/Dennis/' symphonyOfTheNight.txt
Popular scene from Castlevania: Symphony of the Night

Richter: Die monster. You don't belong in this world!
Dennis: It was not by my hand that I am once again given flesh. I was called here by hum
ans who wish to pay me
tribute.
Richter: Tribute!?! You steal men's souls and make them your slaves!
Dennis: Perhaps the same could be said of all religions...
Richter: Your words are as empty as your soul! Mankind ill needs a savior such as you!
Dennis: What is a man? (throws his wine glass to the floor) A miserable little pile of s
ecrets. But enough talk...
Have at you!
```

Change dracula for dennis

4.

```
[alvare12@elvis txtFiles]$ sed 's/can/cannot/g' paragraphs.txt
This file is a bunch of randomly generated paragraphs https://randomword.com/paragraph

He was an expert but not in a discipline that anyone could fully appreciate. He knew how
 to hold the cone just right so that the soft server ice-cream
fell into it at the precise angle to form a perfect cone each and every time. It had tak
en years to perfect and he could now do it without even putting
any thought behind it. Nobody seemed to fully understand the beauty of this accomplishme
nt except for the new worker who watched in amazement.

There was something special about this little creature. Donna couldn't quite pinpoint wh
at it was, but she knew with all her heart that it was true. It
wasn't a matter of if she was going to try and save it, but a matter of how she was goin
g to save it. She went back to the car to get a blanket and
when she returned the creature was gone.

If you cannot imagine a furry humanoid seven feet tall, with the face of an intelligent
gorilla and the braincase of a man, you'll have a rough idea of
what they looked like -- except for their teeth. The cannotines would have fitted better
 in the face of a tiger, and showed at the corners of their wide,
thin-lipped mouths, giving them an expression of ferocity.

Where do they get a random paragraph?" he wondered as he clicked the generate button. Do
 they just write a random paragraph or do they get it
somewhere? At that moment he read the random paragraph and realized it was about random
paragraphs and his world would never be the same...
```

Change can with cannot

5.

```
[alvare12@elvis txtFiles]$ grep -E 'c(an|ou)' paragraphs.txt
He was an expert but not in a discipline that anyone could fully appreciate. He knew how
 to hold the cone just right so that the soft server ice-cream
fell into it at the precise angle to form a perfect cone each and every time. It had tak
en years to perfect and he could now do it without even putting
There was something special about this little creature. Donna couldn't quite pinpoint wh
at it was, but she knew with all her heart that it was true. It
If you can imagine a furry humanoid seven feet tall, with the face of an intelligent gor
illa and the braincase of a man, you'll have a rough idea of
what they looked like -- except for their teeth. The canines would have fitted better in
 the face of a tiger, and showed at the corners of their wide,
```

Looks for any word that starts with “c” and follows by “an” or “ou”.

6.

```
[alvare12@elvis txtFiles]$ sed 's/man/jedi/gI' symphonyOfTheNight.txt
Popular scene from Castlevania: Symphony of the Night

Richter: Die monster. You don't belong in this world!
Dracula: It was not by my hand that I am once again given flesh. I was called here by hu
jedis who wish to pay me
tribute.
Richter: Tribute!?! You steal men's souls and make them your slaves!
Dracula: Perhaps the same could be said of all religions...
Richter: Your words are as empty as your soul! jedikind ill needs a savior such as you!
Dracula: What is a jedi? (throws his wine glass to the floor) A miserable little pile of
 secrets. But enough talk...
Have at you!
```

Change man with jedi even if man was capital

7.

```
[alvare12@elvis txtFiles]$ grep -rl th .
./table.txt
./symphonyOfTheNight.txt
./paragraphs.txt
```

Look for all txt files

8.

Print everyone that is 30 years old

```
[alvare12@elvis txtFiles]$ awk '/30\t/ {print $0}' table.txt
Harold  Fowler  30      Physicist       $48,507
Blake   Stevens 30      Graphic Designer        $132,513
Jenna   Lloyd   30      Botanist        $37,154
Edward  Hunt    30      Medic   $104,501
Ned     Hill    30      Architect       $70,008
Kimberly        Turner  30      Pharmacist      $165,599
Alfred  Perry   30      Economist       $52,599
Mary    Cole    30      Veterinarian    $115,837
Alina   Riley   30      Journalist      $124,299
Jacob   Morgan  30      Composer        $141,492
Reid    Grant   30      Carpenter       $115,694
Julian  Kelly   30      Electrician     $135,725
```

9.

```
[alvare12@elvis txtFiles]$ awk '{FS = "\t"}; {print $1, $2};' table.txt
First Name
Daisy Baker
Sarah Dixon
Caroline Grant
Adrian Mitchell
Ryan Montgomery
Kirsten Owens
Roman Richardson
Marcus Murphy
Roland West
Harold Fowler
Alisa Cole
Byron Bennett
Justin Chapman
Elise Kelley
Valeria Ellis
Ada Reed
Adelaide Mitchell
Julia Smith
Deanna Hall
Ellia Gibson
Blake Stevens
Tess Richards
Derek Hawkins
Jasmine Montgomery
Olivia Brooks
Sofia Kelley
Jenna Lloyd
Steven Turner
Victor Craig
Maria Ross
Myra Stevens
Edward Hunt
Lilianna Russell
Justin Allen
Kate Walker
Honey Thomas
Stuart Johnston
```

Ned Hill
Antony Roberts
Miley Robinson
Tara Morgan
Arthur Adams
Sarah Howard
Kimberly Turner
Alfred Perry
Kevin Warren
Amy Thomas
Henry Thomas
Mary Russell
Maya Howard
Maddie Gibson
Mary Cole
David Ryan
Rosie Higgins
Violet Murphy
Alina Riley
Sienna Elliott
Melanie Bailey
Derek Henderson
Valeria Turner
Freddie Alexander
Stuart Hill
Edwin Andrews
Edith Carroll
Tara Gray
Carlos Ryan
Jacob Morgan
Albert Richardson
Reid Baker
Eric West
Nicholas Hill
Sam Elliott
Alberta Payne
Julia Roberts
Eric Armstrong
David Turner
Adrian Henderson
Violet Reed
Hailey Hall
Thomas Cameron
Alissa Cole
Carina Johnson
Victor Harper
Lyndon Clark
Eric Ross

```
Myra Anderson
Reid Grant
Sawyer Thompson
Sabrina Thompson
Caroline Spencer
Mary Payne
Connie Payne
Violet Parker
Nicholas Brooks
Clark Anderson
Derek Lloyd
Heather Morris
Alina Gray
Adam Thompson
Julian Kelly
```

Print first name and last name

10.

```
[alvare12@elvis txtFiles]$ awk -F '\t' '{if($3 >= 25) {print $4} else {if($5 > 100000) {print $5}}}' table.txt
Occupation
Dancer
Historian
Fashion Designer
Archeologist
Physicist
Chef
Accountant
Lecturer
Producer
Scientist
Graphic Designer
Botanist
Chef
Medic
Architect
Carpenter
Pharmacist
Economist
Photographer
Salesman
Veterinarian
Astronomer
Auditor
Journalist
Lecturer
Veterinarian
Composer
Astronomer
Graphic Designer
Interpreter
Baker
Social Worker
Firefighter
Baker
Carpenter
Mechanic
Cook
Biochemist
Florist
Electrician
```

Print all occupations

11. sed -i ‘s/empty/full/g’ symphonyOfTheNight.txt

```
[alvare12@elvis txtFiles]$ cat symphonyOfTheNight.txt
Popular scene from Castlevania: Symphony of the Night

Richter: Die monster. You don't belong in this world!
Dracula: It was not by my hand that I am once again given flesh. I was called here by humans who wish to pay me
tribute.
Richter: Tribute!?! You steal men's souls and make them your slaves!
Dracula: Perhaps the same could be said of all religions...
Richter: Your words are as full as your soul! Mankind ill needs a savior such as you!
Dracula: What is a man? (throws his wine glass to the floor) A miserable little pile of secrets. But enough talk...
Have at you!
```

12. sed ‘4 s/Dracula/Jersey Devil/g’ symphonyOfTheNight.txt

```
[alvare12@elvis txtFiles]$ sed '4 s/Dracula/Jersey Devil/g' symphonyOfTheNight.txt
Popular scene from Castlevania: Symphony of the Night

Richter: Die monster. You don't belong in this world!
Jersey Devil: It was not by my hand that I am once again given flesh. I was called here by humans who wish to pay me
tribute.
Richter: Tribute!?! You steal men's souls and make them your slaves!
Dracula: Perhaps the same could be said of all religions...
Richter: Your words are as full as your soul! Mankind ill needs a savior such as you!
Dracula: What is a man? (throws his wine glass to the floor) A miserable little pile of secrets. But enough talk...
Have at you!
```

13.grep this *

```
[alvare12@elvis txtFiles]$ grep this *
paragraphs.txt:any thought behind it. Nobody seemed to fully understand the beauty of this accomplishment except for the new worker who watched in amazement.
paragraphs.txt:There was something special about this little creature. Donna couldn't quite pinpoint what it was, but she knew with all her heart that it was true. It
symphonyOfTheNight.txt:Richter: Die monster. You don't belong in this world!
```

14.grep -o ‘\ba\w*[sn]\w*d\b’ paragraphs.txt

```
[alvare12@elvis txtFiles]$ grep -o '\ba\w*[sn]\w*d\b' paragraphs.txt
and
and
and
and
and
and
and
and
```

15. awk ‘/Historian/{print $0}’ table.txt

```
[alvare12@elvis txtFiles]$ awk '/Historian/ {print $0}' table.txt
Daisy   Baker   22      Historian       $61,003
Kirsten Owens   29      Historian       $143,693
Roman   Richardson      18      Historian       $156,094
```

16.

```
[alvare12@elvis chmod]$ tree -anp
[drwx------]  .
â"â"€â"€ [drwx----wx]  fish
â",Â Â  â""â"€â"€ [drwxrwx---]  two
â",Â Â      â""â"€â"€ [---x-w--wx]  fish
â""â"€â"€ [drwx------]  one
    â"â"€â"€ [--wx-w---x]  blue
    â""â"€â"€ [-r--r--r--]  red

3 directories, 3 files
```

17.

```
[alvare12@elvis midterm]$ awk 'BEGIN{for(i=5;i<=10;i++){if(i==7){print "That\u0027s my l
ucky number!"}else{print i}}}'
awk: cmd. line:1: warning: escape sequence `\u' treated as plain `u'
5
6
Thatu0027s my lucky number!
8
9
10
```

18.

```c
#include <stdio.h>
#include <string.h>

#define MAX_Length 100

int main() {
    char input[MAX_Length];
    int c = 0;
    int l;

    printf("Enter input: ");
    fgets(input, MAX_Length, stdin);

    l = strlen(input);

    for (int i = 0; i < l - 1; i++) {
        if (input[i] == 'o' && input[i+1] == 'f') {
            c++;
        }
    }

    printf("Number of 'of' occurrences: %d\n", c);

    return 0;
}
```

19.

```c
#include <stdio.h>

#define ARRAY_SIZE 10

int main() {
    int array[ARRAY_SIZE] = {1, 2, 3, 4, 5, 6, 7, 8, 9, 10};
    int *ptr = &array[ARRAY_SIZE - 1];

    for (int i = 0; i < ARRAY_SIZE; i++) {
        printf("%d\n", *ptr);  // Print value pointed to by pointer
        ptr--;  // decrement
    }

    return 0;
}
```

```
[alvare12@elvis cFiles]$ ./a.out pointer.c
10
9
8
7
6
5
4
3
2
1
```

20.

```
[alvare12@elvis cFiles]$ gcc main.c rowan.c university.c college.h
[alvare12@elvis cFiles]$ ./a.out
Funtions from rowan.c and university.c
Hello to all Rowan University students!
from the university!

Welcome to the College!
```

```
#ifndef COLLEGE_H
#define COLLEGE_H

        void rowan();
        void university();
        char message[] = "Welcome to the College!";



#endif /* COLLEGE_H*/
```

```
#include <stdio.h>


        void *rowan(){

        printf("Hello to all Rowan University students!\n");

        return 0;


        }
```

```c
#include <stdio.h>

        void *university(){

        printf("from the university!\n");


        }
```

```c
#include <stdio.h>
#include "college.h"



void rowan();
void university();

int main(){


        printf("Funtions from rowan.c and university.c\n");
         rowan();
         university();

        printf("\n%s\n", message);



        return 0;



        }
```